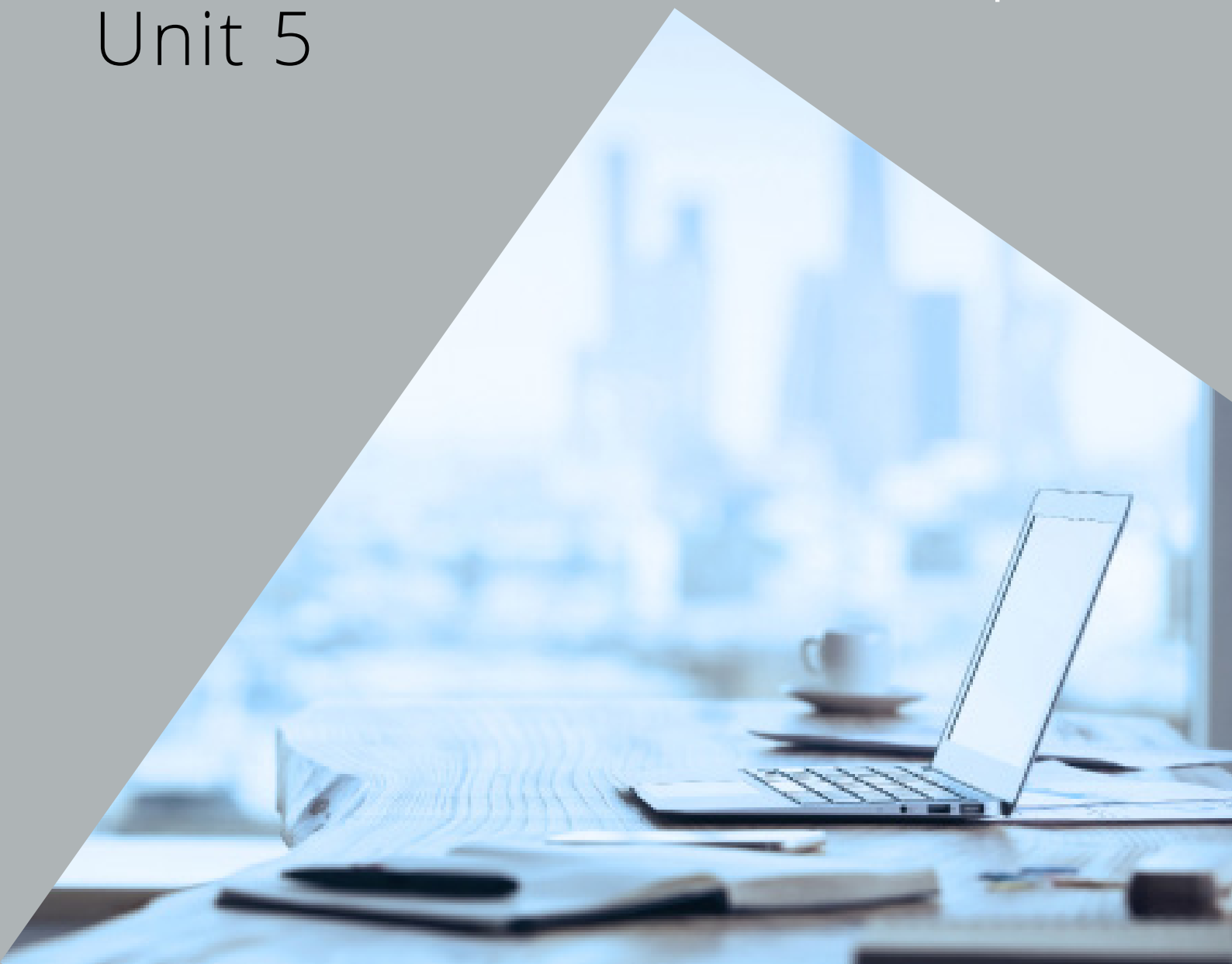


vision**2**learn

The Principles of Cyber Security Level 2

Hints
and
Tips

Unit 5



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Hints and Tips Unit 5

The information in this guide will be helpful to you when you are ready to complete your assessment for the following unit:

Unit 5: Understand methods of maintaining cyber security

Section 1 – Understand preventative methods used to maintain cyber security

Q1. In the table below, **describe** two routine methods used to maintain cyber security.

This question is quite straightforward. You simply need to select two of the methods covered in the course materials and give a description of how they are used to maintain cyber security. This is a describe question so try to provide good detail in your answer.

Q2. In the box below, **explain** what is meant by vulnerability testing.

This is an explain question so you should provide good detail in your answer. You should give examples of the different ways of testing in your response.

Q3. In the box below, **explain** what is meant by penetration testing.

This is another explain question so try to be as detailed as possible in your response. What does penetration testing do, when should it be used?

Q4. In the box below, **explain** what is meant by security updates.

For this question, as well as explaining what is meant by security updates you could use your own computer system to give an example. What does your computer do in relation to security updates, what are they for, when should they be installed, why are they important?

Q5. In the box below, **explain** what is meant by 'patching'.

This is another explain question so try to provide good detail in your answer. Think about what patching is, when it is used, why is it used, are there any issues in relation to patching? Is this something that you have had to do on your own computer system?

Q6. In the box below, **explain** why cyber security information and maintenance records should be accurate and up to date.

For this question you should consider what the benefits of making sure cyber security information is accurate and up to date. You also need to consider the consequences if it is not up to date.

Section 2 – Understand the principles of user access control

Q1. In the box below, **define** what is meant by user access control.

For this question you simply need to give a definition of the term 'user access control'. Use the course material or your own research to help you with this but ensure that your response is in your own words.

Q2. In the table below, **identify** two methods that can be used to restrict user access.

This question is quite straightforward and you simply need to identify how malicious software infects a system. This is an identify question so you can list your responses.

Q3. You are the **IT administrator** for an organisation with 100 employees. There are 4 senior managers with overall control of the business and 8 team managers. You have been asked to **set up a user access control system** for all employees. In the table below, explain the steps that you would take to create this system.

You must consider the following in your response:

- Usernames and password allocation
- Administration accounts and administrator permissions
- Setting up file/folder permissions for you as overall admin, Senior managers and team managers.

For this question you need to read the scenario carefully and then follow a step by step approach to setting up the user access control system. Make sure that you include the following in your response.

The use of usernames and password allocation, and how administrator level access can block users from installing unauthorised applications/software and making system changes that could compromise security. The following website will assist you with this. [Designing and configuring a system of access levels \(looker.com\)](https://www.looker.com/blog/designing-and-configuring-a-system-of-access-levels)

Section 3 – Understand the use of firewalls

Q1. In the box below, **define** the term 'firewall'.

This question is straightforward and you simply need to state what a firewall is.

Q2. In the box below, **describe** the key functions of a firewall.

This is a describe question so try to be as detailed as possible in your response. What does it do, what is its purpose, how does it protect your systems?

Q3. In the box below, **define** what is meant by network traffic.

This question is quite straightforward and you simply need to state what network traffic is.

Q4. In the table below, **explain** the advantages and disadvantages of using a firewall to protect from threats. You must give two advantages and two disadvantages.

For this question think of two of the advantages that a firewall gives you and two disadvantages. Try to provide good detail in your response as this is an explain question.